

Research article

A new blockchain oriented deep reinforced active learning framework for robust IoT intrusion detection

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ABSTRACT

In IoT networks, traffic data are high-dimensional, complex, and highly imbalanced. Minority attack classes are often underrepresented degrading the detection performance. In addition, manual labeling of large-scale IoT traffic is costly, and network behavior continuously changes over time. To overcome these challenges, this paper presents an adaptive, secure and explainable IoT intrusion detection framework. First, a generative AI based data balancing method is applied using the real-valued non-volume preserving model. It generates realistic minority-class samples while preserving the original data distribution. Then, a hybrid Deep Dual-Attention Network (DDANet) is proposed. It combines a Deep Neural Network (DNN) with a Dual Attention Network (DANet). The DNN captures global nonlinear features. The DANet focuses on important channel and spatial features. This improves feature learning and intrusion detection accuracy. Next, Active DDANet (ADDANet) integrates a monte-carlo based active learning strategy. It selects informative samples using predictive uncertainty to reduce redundant labeling and improves learning efficiency. Reinforced DDANet (RDDANet) further integrates an actor-to-critic reinforcement learning mechanism. It enables adaptive optimization in dynamic IoT environments and stabilizes training. A blockchain layer is also integrated. It ensures secure logging, decentralized trust, and tamper-resistant storage of detection results. This increases transparency and reliability in distributed IoT systems. Experimental results show that DDANet improves accuracy, recall, and F1-score by 3.33%, precision by 3.30%, Cohen's Kappa (CK) and Mathew's Correlation coefficient (MCC) by 4.60%, and Precision Recall Area Under the Curve (PR-AUC) by 3.16% over baseline models. It also reduces Log Loss (LL) by 34.38% and Hamming Loss (HL) by 30.00%. ADDANet further improves accuracy, recall, and F1-score by 3.23%, CK and MCC by 3.30%, and reduces LL by 38.10% and HL by 42.86% over DDANet. RDDANet achieves similar accuracy improvement of 3.23%, increases CK and MCC by 4.40%, and provides the highest LL reduction of 42.86% and HL reduction of 42.86%. The framework is validated using 10-fold cross-validation to ensure robustness and generalization. Confidence interval analysis and paired *t*-test confirm statistical significance. Finally, integrated gradients based explainable AI is applied to provide feature attribution scores and improve model transparency. The proposed framework provides an adaptive, secure, explainable, and high-performing IoT intrusion detection solution.

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1. Background

The Internet of Things (IoT) is a massive distributed computing paradigm where physical entities are integrated with sensors, actuators and communication devices [1]. Such devices communicate with one another and with either cloud or edge platforms to provide intelligent services [2]. IoT networks facilitate various applications, namely smart healthcare, smart transportation system, smart grids, industrial control systems, and environmental monitoring. The IoT architecture generally consists of perception, network, and application layers all producing and performing enormous amounts of heterogeneous data [3]. This distributed and layered nature will require constant surveillance and data management control systems.

The Intrusion Detection Systems (IDS) are now a critical element of IoT infrastructures as far as security is concerned. The IoT environments are unlike the traditional enterprise networks, which are extremely dynamic and resource-constrained [4]. The current IDS models are becoming more based on Machine Learning (ML) and Deep Learning (DL) models to examine traffic flows, derive discriminating features, and categorize normal and malicious traffic [5]. Such methods of data modeling can capture nonlinearities of network traffic of high dimensions. In addition to that, adaptive learning mechanisms can allow the IDS models to match the evolving traffic pattern and fresh threat, making them suitable to large-scale IoT applications.

An area of significant concern is security of distributed IoT system through trust, transparency, and integrity. The blockchain technology introduces a decentralized ledger system where the transactions are verified through consensus mechanisms and cryptographic hash. The last block links with the next block forming an irredeemable chain of records. This design will ensure that no unauthorized alteration of the stored data is made and enhance the levels of audit. IoT blockchain can also be employed to deliver secure authentication of the devices, secure exchange of data, and resist tampering logging of the network events.

Integration of intelligent IDS and blockchain-based trust management is also an important phenomenon in the studies of the IoT cybersecurity. The accuracy and flexibility of the detection can be enhanced with the help of artificial intelligence (AI), and the data integrity and decentralized coordination of the nodes can be enhanced with the help of the Blockchain. This integration promotes the creation of safe, open and scalable IoT systems. With the further development of IoT applications in critical infrastructure fields, the integration of sophisticated analytics and distributed ledger technologies is one of the main ways to create resilient and reliable network environments.

Recent researches concerning the use of IDSs to operate in the IoT environments are examined in this study and they include data imbalance management, data learning architecture, and intelligent strategies of learning. In recent works, the idea of using Generative AI (GAI) to data balance IoT intrusion detection has been discussed multiple times. The initial methods based on GAN involve adversarial training to synthesize samples of minority classes in order to address the problem of class imbalance [2]. Whereas, these approaches enhance the detection rates of rare attacks, adversarial training usually experiences mode collapse and unstable training, which deteriorates with the network traffic distributions that are difficult to learn and detect, and in general, hard to predict and foresee [3]. Other methods of balancing incorporating Auto Encoders (AE) have been examined in which minority samples of classes are reconstructed by use of latent representations [5]. Such models have the ability to reproduce latent patterns, but occasionally do not maintain high-level discriminative features, resulting in poor generalization of imbalanced IoT data.

Recently, in the intrusion detection modeling, K-Nearest Neighbor (KNN) and the Logistic Regression (LR) are classical classifier and have been analyzed in terms of their performance against IoT datasets [6,7]. Such models are computationally effective, little preprocess is needed, and they have explicit decision boundaries, which make them easier to interpret. Nonlinear relationship and high dimensional dependence in dynamic network flows, however, are difficult to capture using them. They can even break down further in the case of noisy features or grossly imbalanced distributions of attacks since tree-based splits are sensitive to majority classes.

ShuffleNet and other compact convolutional based architectures are also implemented to be used in lightweight IDS in a resource constrained IoT environment [8]. ShuffleNet is channel shuffled and grouped convolution making it to be a good fit to edge devices. However, its less depth features and condensed feature maps can restrict its ability to observe the minor and uncommon attacks. Deep Neural Network (DNNs) are still popular because of their high representation learning capacity and possibility to extract deep feature hierarchies of raw traffic features [9]. The traditional DNNs, though effective, treat the features in the same way and fail to account on the relative importance of the discriminative attributes.

In order to overcome this drawback, Dual Attention Networks (DANet) employ spatial and channel attention to highlight informative areas and feature channels [10]. The attention mechanisms enhance refining of features and increase the concentration of model on important traits in traffic. Nevertheless, independent application of such architectures usually fails to explore complementary learning between global nonlinear extraction and attention-based refinement. Consequently, the available models might fail to reflect feature interdependencies and hence deliver suboptimal detection performance, especially in a highly imbalanced and heterogeneous IoT traffic environment.

Strategies of Active Learning (AL) which include entropy, uncertainty and diversity have been explored to eliminate manual labelling in intrusion detection. Entropy based sampling is more concerned with information gain and is likely to pick noisy samples [11]. Uncertainty sampling concentrates on the questionable cases but can not concentrate on the diversity in representations [12]. Diversity-based selection is a solution to such problems because it provides a wide coverage of features [13]. Although such innovations exist, the most research has used single-criterion selection, and very little incorporates probabilistic sampling with diversity, which constrains the informativeness of the sample. In addition, the majority of the currently available AL frameworks suppose that data distributions are fixed, and they do not take into account the changing attack patterns in the IoT environment. This can create a limitation in query strategy in case new or uncommon attack behavior arises. Hence, more adaptive and distribution-sensitive AL

mechanisms that can be able to balance uncertainty estimation and representative data coverage are still required.

Adaptive intrusion detection has recently been handled by Reinforcement Learning (RL) techniques. The policy learning has been stabilized in changing IoT environments with the use of Proximal Policy Optimization (PPO) [14]. Although PPO offers clipped updates and less policy variance, it has to be hyperparameter-tuned carefully and is too conservative when facing high variance of policies [15]. Directly optimizing the rewards is also examined by the policy gradient-based methods such as the REINFORCE, which uses expected rewards as a part of its criterion but has high variance and is slow to converge to a solution in practice [16]. Value functions have been learnt with Deep Q-Learning (DQL) to perform sequential decision making in the context of IDS situations, which demand discrete action space and target networks, making it difficult to apply to continuous or large action space situations [17]. Despite the above approaches yielding valuable insights, they tend to look at isolated parts of the intrusion detection pipeline like data balancing, model architecture or learning strategy. The need exists to have framework that combine the issues of class imbalance, hybrid feature learning, adaptive reinforcement policies, and intelligent sample selection with strict statistical validation in IoT intrusion detection.

1.1. Research gaps

The key research gaps identified in the existing literature are summarized as follows:

- Existing data balancing techniques such as oversampling, undersampling, and basic GAN-based generation often fail to preserve the true distribution of IoT traffic data. Many generative models introduce noise or mode collapse, which reduces the quality and diversity of synthetic attack samples. There is a lack of stable and distribution-preserving generative approaches for realistic minority class augmentation.
- Most existing intrusion detection models rely on single-stream DL architectures. These models often focus on either global feature extraction or attention-based refinement, but rarely combine both effectively. Limited research has explored parallel hybrid architectures that simultaneously capture deep nonlinear patterns and feature-level attention dependencies in IoT traffic.
- Traditional supervised learning approaches require large volumes of labeled data. However, labeling IoT network traffic is expensive and time-consuming. Current AL strategies are often uncertainty-driven only and do not adequately consider diversity or probabilistic uncertainty estimation. There remains a need for more reliable and informative sample selection mechanisms.
- Many RL-based IDS models employ value-based methods or high-variance policy gradient techniques. These approaches may suffer from unstable convergence and poor adaptability in dynamic IoT environments. The integration of stable Actor-to-Critic (A2C) optimization techniques for adaptive intrusion detection remains underexplored.
- Although blockchain has been proposed for IoT security, most existing implementations focus primarily on data storage or access control. Limited work has examined its integration with intelligent intrusion detection frameworks for secure logging, transparent decision validation, and decentralized trust reinforcement. A comprehensive fusion of blockchain with adaptive IDS mechanisms is still lacking.
- Several studies report results using a single train-test split, which may produce optimistic or biased performance estimates. Limited research emphasizes robust validation strategies for consistent and generalized performance assessment in IoT intrusion detection.
- Many existing works present only mean performance metrics without reporting statistical dispersion measures. The absence of confidence interval analysis limits understanding of model stability, reliability, and performance consistency across multiple runs or data partitions.
- Comparative studies often lack formal statistical hypothesis testing. Improvements are frequently reported without verifying their statistical significance. The use of paired statistical tests, such as the paired *t*-test, remains insufficient in validating performance gains in IDS research.
- Despite the high accuracy of DL-based IDS models, interpretability remains limited. Many studies do not incorporate explainable AI techniques to justify model predictions. The absence of feature attribution analysis reduces transparency and trust in security-critical IoT applications.

1.2. Major contributions

To overcome these research gaps, the main contributions of this paper are summarized as follows:

- A synthetic data generation framework based on GAI is developed using the Real-Valued Non-Volume Preserving (Real-NVP) technique for class balancing. This contribution is important because IoT intrusion datasets are highly imbalanced, which degrades detection performance for minority attack classes. Real-NVP preserves the underlying data distribution while generating realistic samples, thereby improving model fairness and robustness.
- A novel Deep Dual-Attention Network (DDANet) is proposed as a parallel hybrid architecture that integrates a DNN with a DANet. This design is important because IoT traffic features are high-dimensional and complex. The DNN captures global nonlinear feature relationships, while the dual attention mechanism emphasizes both channel-wise and spatial feature importance, leading to more discriminative intrusion detection.
- An Active DDANet (ADDANet) is introduced by incorporating a Monte-Carlo (MC)-based AL strategy. This contribution is significant because labeling large-scale IoT traffic data is costly and time-consuming. The MC sampling mechanism estimates predictive uncertainty through multiple stochastic forward passes, selects informative and diverse samples, and improves learning efficiency while reducing redundant annotations.

- A Reinforced DDANet (RDDANet) is developed using an A2C RL technique. This is important because IoT environments are dynamic and continuously evolving. The A2C mechanism reduces policy variance, stabilizes convergence, and enables adaptive optimization of detection strategies under changing network conditions.
- A blockchain-based implementation is integrated into the framework to ensure secure logging and decentralized trust management. This contribution is essential because IoT systems operate in distributed environments where data integrity and transparency are critical. The blockchain layer provides tamper-resistant storage, cryptographic validation, and reliable auditability of intrusion detection outcomes.
- A rigorous experimental validation strategy is adopted using 10-Fold Cross-Validation (10-FCV). This contribution is important because single train-test splits produce biased results. 10-FCV ensures robustness, reduces overfitting risk, and provides stable performance estimation across different data partitions.
- Confidence interval analysis is performed to quantify the reliability of the reported performance metrics. This is significant because it provides statistical bounds around the mean results and demonstrates the consistency and generalization capability of the proposed models.
- A paired *t*-test statistical analysis is conducted to verify the significance of performance improvements over comparative models. This contribution is important because it validates that the observed improvements are not due to random variation but are statistically meaningful.
- An explainable AI analysis is performed using the Integrated Gradients (IGs) technique. This is essential because DL models often operate as black boxes. IGs provide feature attribution scores, improve transparency, and enhance trust in intrusion detection decisions within critical IoT environments.

The rest of the paper is organized into different sections. [Section 2](#) presents the proposed methodology, the experimentation results are discussed in [Sections 3](#), and [4](#) concludes this work and sets future directions.

2. Proposed IoT intrusion detection framework

The proposed methodology presents an adaptive, secure and explainable framework to IoT intrusion detection, which involves sophisticated data processing, hybrid DL models, and secure systems design. It starts with a generative data balancing method to solve the issue of class imbalance, which is followed by the creation of the new hybrid models incorporating attention mechanisms, dense connections, RL, and AL methods. Both models are intended to be able to define complicated dependencies in the traffic of an IoT network. Moreover, blockchain implementation guarantees the protection and integrity of data processing among distributed IoT devices. The framework is tested with high accuracy of a rigorous statistical techniques and explainability approach with high detection accuracy and explainable features in importance.

2.1. Network traffic dataset for web intrusion analysis

The experiments are carried out on a flow-based intrusion detection dataset based on the CIC-IDS-2017 benchmark, namely, the web attack traffic of Thursday morning [18]. The experiments are based on a real world network of the enterprise setting, and involve both normal user traffic and a set of attacks based on web. Packet capture and flow analysis tools are applied to raw packet traces to produce bidirectional network flows. Every flow is described by 79 features that describe statistical, temporal and behavioral properties of network traffic. These characteristics characterize flow-duration, packet and byte counts, inter-arrival time behavior and information at protocol level. These elaborate flow representations enable the learning model to learn these finer details in distinguishing normal traffic and the various attack patterns.

This study cuts down the initial dataset to a sample of 50,000 cases which ensures effective model training and uniform analysis of the experiment. The sampling is applied in a manner that the original distributions of the classes will be retained to ensure realistic proportions of the traffic are maintained hence not to be biased in the sampling. There are four categories in the data set and one of them is the normal traffic which is represented as the *BENIGN* and the other three categories are represented as the *Web Attack-Brute Force*, *Web Attack-SQL Injection* and *Web Attack-XSS*. This system enables a multiclass disposition task with the intention of identifying web application attacks and realistically viable performance measurement in realistic network environments.

2.2. Data preprocessing

The dataset goes through a sequence of preprocessing steps before the sample selection can be done, so that the quality of data and compatibility with a learning model can be guaranteed. These measures involve duplicating row removal, labeling of categories and normalizing features. Each step is illustrated in [Fig. 1](#) and described below.

2.2.1. Elimination of duplicate instances

The dataset has 170,366 instances in the beginning. In the process of preprocessing, 6066 rows that have been duplicated are eliminated in order to avoid redundancy and bias in model training [19]. Following this step, there are 164,300 unique instances in the dataset. Duplicates should be avoided in order to make sure that the model is learning different samples and is not overfitting to recurring patterns. The operation can be mathematically represented as:

$$D_{\text{unique}} = D - D_{\text{duplicate}} \quad (1)$$

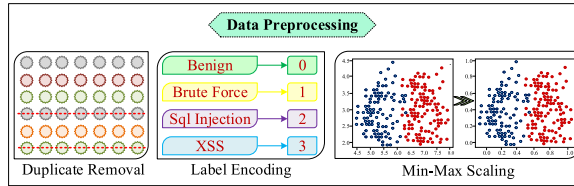


Fig. 1. Data preprocessing steps for CIC-IDS-2017 dataset.

where D is the original dataset, $D_{\text{duplicate}}$ represents all duplicate rows, and D_{unique} is the resulting dataset after removal. This step guarantees that only unique and meaningful instances are used for further processing.

2.2.2. Categorical label transformation

Categorical class labels, representing normal and web attack traffic types, are transformed into numerical values using label encoding [20]. This conversion is necessary for the classifier to interpret class information in numeric format. Each class, including *BENIGN*, *Web Attack-Brute Force*, *Web Attack-SQL Injection*, and *Web Attack-XSS*, is assigned a unique integer code. Mathematically, label encoding can be expressed as:

$$y_i = f(c_i) \quad (2)$$

where c_i is the original categorical label of the i th instance, f is the encoding function that maps each class to a unique integer, and y_i is the encoded numeric label. This step ensures that class information is machine-readable and suitable for model training.

2.2.3. Feature normalization using min-max scaling

After label encoding, all feature values are normalized using min-max scaling. This transforms each feature into a fixed range, typically $[0, 1]$, ensuring uniform contribution of features during model training [21]. Normalization improves convergence speed and overall model performance by preventing features with larger ranges from dominating the learning process. The min-max scaling is computed as:

$$x'_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (3)$$

where x_{ij} is the original value of the j th feature for the i th instance, $\min(x_j)$ and $\max(x_j)$ are the minimum and maximum values of the j th feature across all instances, and x'_{ij} is the scaled feature value. This step ensures that all features contribute proportionally during model training and improves learning stability.

2.3. Synthetic data generation using real-valued non-volume preserving for class balancing

The data is highly skewed in terms of classes. The majority are of the *BENIGN* type with 98.72 percent and the other minority type of the classification is *Web Attack-Brute Force*, *Web Attack-XSS* and *Web Attack-SQL Injection* with very low number of samples. When trained on such skewed data, the DL models will favor the majority class and fail to detect the infrequent attacks. In order to solve this issue we are using the generative model of RealNVP to artificially upsample minority groups and obtain a balanced dataset. RealNVP is a type of normalizing flow model that learns a bijective mapping between a simple latent distribution and complex data distributions [22]. It transforms standard Gaussian noise, $z \sim \mathcal{N}(0, I)$, into synthetic samples that mimic the statistical properties of the original data:

$$x = f_{\theta}(z), \quad (4)$$

where x represents a generated sample and f_{θ} is an invertible function parameterized by θ . The invertibility of the transformation ensures that no information is lost and that the generated samples preserve the structure of the original feature space, as shown in Fig. 2. The training objective maximizes the log-likelihood of the observed data, which can be written as:

$$\log p_X(x) = \log p_Z(f_{\theta}^{-1}(x)) + \log \left| \det \frac{\partial f_{\theta}^{-1}(x)}{\partial x} \right|. \quad (5)$$

Here, the first term encourages the model to match the latent distribution to real data, while the second term, the log-determinant of the Jacobian, accounts for the change in volume during transformation. Each minority class is modeled independently, and the number of training epochs is adapted according to the class size to prevent overfitting and ensure adequate learning from scarce samples. After training, synthetic samples are generated by inverting the learned transformations:

$$\tilde{x}_i = f_{\theta}^{-1}(z_i), \quad i = 1, \dots, N_{\text{needed}}. \quad (6)$$

where N_{needed} is the number of samples required to match the majority class. These generated samples are then combined with the original dataset to create a balanced dataset:

$$X_{\text{balanced}} = [X_{\text{original}}; \tilde{X}], \quad y_{\text{balanced}} = [y_{\text{original}}; \tilde{y}]. \quad (7)$$

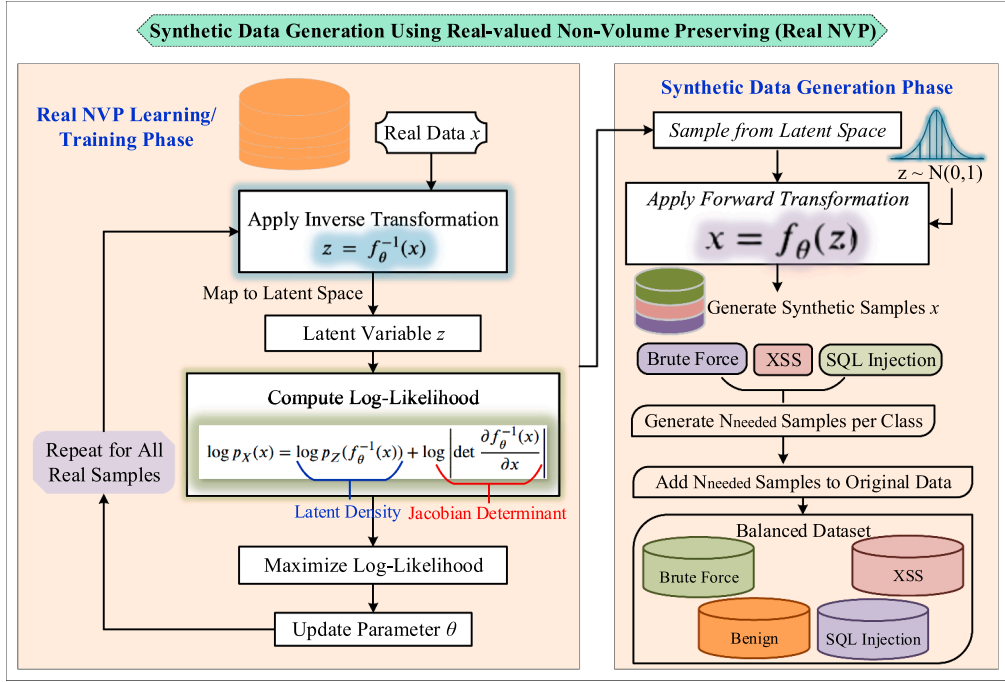


Fig. 2. Synthetic data generation using real-valued non-volume preserving.

This synthetic data augmentation ensures that each class has comparable representation, which allows the model to learn discriminative patterns for both common and rare classes. Moreover, using RealNVP preserves the correlation and distributional characteristics of the original features, reducing the risk of introducing artifacts in the synthetic data. By balancing the classes, the model achieves higher sensitivity to minority attack categories and improved overall detection performance, while maintaining stability and generalization across the dataset. RealNVP-based balancing provides a robust and scalable solution for class imbalance in tabular network intrusion datasets. It addresses the challenges of extreme class skew without discarding valuable information from the majority class, and it enhances the learning capability of ML and DL models for multiclass web attack detection.

2.4. Proposed deep dual-attention network (DDANet)

In this study, we propose a novel hybrid model DDANet for multiclass web attack detection. DDANet combines the strengths of a DANet and a DNN to capture global feature dependencies. The hybrid architecture is designed to exploit complementary features from both networks, enhancing the model's ability to discriminate between closely related attack categories. The first component of DDANet is the DNN, which is particularly effective in learning sequential dependencies in tabular network traffic data [9]. Formally, given an input $x \in \mathbb{R}^d$, the DNN applies a series of transformations with dropout regularization:

$$h_1 = \tanh(W_1 x + b_1), \quad h_1^{dp} = \text{Dropout}(h_1) \quad (8)$$

$$h_2 = \tanh(W_2 h_1^{dp} + b_2), \quad h_2^{dp} = \text{Dropout}(h_2) \quad (9)$$

$$y_{\text{dnn}} = \text{softmax}(W_o h_2^{dp} + b_o) \quad (10)$$

where W_i and b_i represent learnable weights and biases. The DNN output y_{dnn} encodes class-specific information, which is useful for distinguishing subtle patterns among attack types [23]. The second component is the DAN, which emphasizes both channel-wise and feature-level interactions. The input x is first passed through multiple dense layers to extract hierarchical representations:

$$h^{(l)} = \tanh(W^{(l)} h^{(l-1)} + b^{(l)}), \quad l = 1, \dots, L \quad (11)$$

To implement channel-wise attention, the network generates an attention vector α using a sigmoid activation [10]:

$$\alpha = \sigma(W_a h^{(L)} + b_a) \quad (12)$$

The feature representation is then modulated by the attention weights:

$$h_{\text{att}} = h^{(L)} \odot \alpha \quad (13)$$

where $\odot$ denotes element-wise multiplication. This mechanism allows the network to focus on informative features while suppressing irrelevant ones, improving discrimination among attack classes, as illustrated in Fig. 3. DDANet integrates the outputs of the DNN

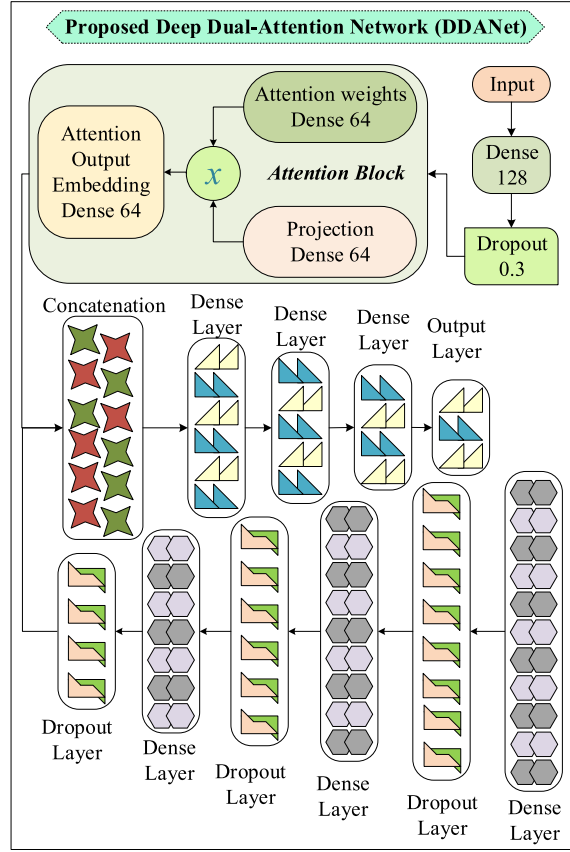


Fig. 3. Working of proposed deep dual-attention network.

and the DANet in a parallel hybrid fusion layer. Let f_{dnn} and f_{dan} represent the feature vectors from each network. The fused feature vector is constructed as:

$$f_{\text{hybrid}} = [f_{\text{dnn}}; f_{\text{dan}}] \quad (14)$$

This concatenation captures complementary information: cues from the DNN and attention-modulated feature importance from DAN. The resulting vector f_{hybrid} serves as input to a meta-classifier. The meta-classifier is a multi-layer perceptron with decreasing layer dimensions to progressively refine the fused representation:

$$h_1^{\text{meta}} = \text{ReLU}(W_1 f_{\text{hybrid}} + b_1) \quad (15)$$

$$h_2^{\text{meta}} = \text{ReLU}(W_2 h_1^{\text{meta}} + b_2) \quad (16)$$

$\vdots$

$$y_{\text{dap}} = \text{softmax}(W_o h_L^{\text{meta}} + b_o) \quad (17)$$

The output y_{dap} provides the final class probabilities. This hierarchical design allows DDANet to integrate low-level and high-level features effectively, improving the model's sensitivity to complex attack patterns. The training of DDANet is performed in stages. First, the DNN and DAN are trained independently on the balanced dataset. Their intermediate outputs are extracted as feature embeddings. Next, these embeddings are concatenated and fed into the meta-classifier. This staged training ensures that each network learns specialized representations before joint fusion. The hyperparameters used in this model are listed in Table 1. Mathematically, the training objective minimizes the categorical cross-entropy loss over the final output:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (18)$$

where N is the number of training samples, C is the number of classes, $y_{i,c}$ is the one-hot label, and $\hat{y}_{i,c}$ is the predicted probability, as describe in Algorithm 1. DDANet combines the strengths of both attention-driven and dense layers-based architectures. The dual attention mechanism highlights informative features, while the DNN module captures subtle patterns in the data. The hybrid fusion layer and meta-classifier ensure that these complementary features are effectively integrated. This design improves the model's

Table 1
Hyperparameter settings for the proposed deep dual-attention network.

Hyperparameter	Value
Output Classes	4
DNN Hidden Dimension	32
DNN Dropout Rate	0.6
DAN Dense Layers	[128, 64, 64, 32, 128]
DAN Dropout Rate	0.4
DAN Attention Activation	Sigmoid (channel-wise)
Meta Dense Layers	[256, 128, 64, 32]
Meta Activation	ReLU
Output Activation	Softmax
Optimizer	Adam
Learning Rate	0.03
Batch Size	128
Epochs	5
Regularization	L2 (1e-3) for DNN
Noise Layer	Gaussian Noise (0.1) in DNN

capability to detect diverse web attacks and handle complex tabular network traffic patterns, making it suitable for large-scale intrusion detection tasks.

Algorithm 1 Information flow in deep dual-attention network.

Require: Balanced training data $X_{\text{train}}, Y_{\text{train}}$, test data X_{test}

Ensure: Predicted class probabilities y_{dap}

- 1: Stage 1: DNN Feature Extraction
 - 2: Apply dense layers with dropout to learn patterns:
 - 3: $h_1^{\text{dnn}} = \tanh(W_1 X + b_1)$
 - 4: $h_1^{\text{dnn-dp}} = \text{Dropout}(h_1^{\text{dnn}})$
 - 5: $h_2^{\text{dnn}} = \tanh(W_2 h_1^{\text{dnn-dp}} + b_2)$
 - 6: $h_2^{\text{dnn-dp}} = \text{Dropout}(h_2^{\text{dnn}})$
 - 7: Output feature vector: $f_{\text{dnn}} = \text{softmax}(W_o h_2^{\text{dnn-dp}} + b_o)$
 - 8: Stage 2: DAN Feature Extraction
 - 9: Extract hierarchical features using multiple dense layers:
 - 10: $h^{(0)} = X$
 - 11: **for** $l = 1$ to L **do**
 - 12: $h^{(l)} = \tanh(W^{(l)} h^{(l-1)} + b^{(l)})$
 - 13: **end for**
 - 14: Compute channel-wise attention: $\alpha = \sigma(W_a h^{(L)} + b_a)$
 - 15: Apply attention to modulate features: $f_{\text{dan}} = h^{(L)} \odot \alpha$
 - 16: Stage 3: Feature Fusion and Meta Layers
 - 17: Concatenate DNN and DAN features to form hybrid representation:
 - 18: $f_{\text{hybrid}} = [f_{\text{dnn}}; f_{\text{dan}}]$
 - 19: Pass hybrid features through multi-layer perceptron to refine representations:
 - 20: $h_1^{\text{meta}} = \text{ReLU}(W_1^{\text{meta}} f_{\text{hybrid}} + b_1^{\text{meta}})$
 - 21: $h_2^{\text{meta}} = \text{ReLU}(W_2^{\text{meta}} h_1^{\text{meta}} + b_2^{\text{meta}})$
 - 22: $h_3^{\text{meta}} = \text{ReLU}(W_3^{\text{meta}} h_2^{\text{meta}} + b_3^{\text{meta}})$
 - 23: $h_4^{\text{meta}} = \text{ReLU}(W_4^{\text{meta}} h_3^{\text{meta}} + b_4^{\text{meta}})$
 - 24: Compute final class probabilities: $y_{\text{dap}} = \text{softmax}(W_o^{\text{meta}} h_4^{\text{meta}} + b_o^{\text{meta}})$
 - 25: Return predicted probabilities: $\hat{y} = y_{\text{dap}}$
-

2.5. Proposed active deep dual-attention network (ADDANet)

The proposed ADDANet integrates a MC-based AL framework to enhance learning efficiency with limited labeled data. The dataset is first divided into three disjoint subsets: a fixed test set, a small initial labeled set, and a large unlabeled pool. Specifically, 20% of the data is reserved for testing, while the remaining 80% is split into labeled and unlabeled subsets, reflecting realistic scenarios where data annotation is expensive or time-consuming. Let the labeled set be denoted as D_L and the unlabeled pool as D_U . Initially, only a fraction of samples is available for supervised training, and the model is first trained on D_L using sparse categorical cross-entropy

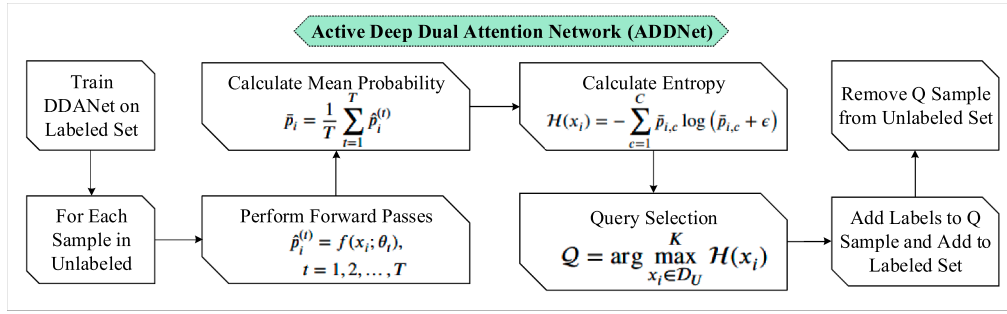


Fig. 4. Working flow of proposed active deep dual-attention network.

loss.

MC dropout is employed to estimate model uncertainty. During inference, dropout layers remain active, and the network performs T stochastic forward passes [24]. For each unlabeled sample $x_i \in D_U$, the predictive probabilities for each forward pass are calculated as

$$\hat{p}_i^{(t)} = f(x_i; \theta_t), \quad t = 1, 2, \dots, T \quad (19)$$

where θ_t represents the model parameters sampled at the t th forward pass. The mean predictive probability across all MC passes is then computed as

$$\bar{p}_i = \frac{1}{T} \sum_{t=1}^T \hat{p}_i^{(t)} \quad (20)$$

The predictive uncertainty of each sample is quantified using Shannon entropy:

$$\mathcal{H}(x_i) = - \sum_{c=1}^C \bar{p}_{i,c} \log(\bar{p}_{i,c} + \epsilon) \quad (21)$$

where C denotes the number of classes and ϵ is a small constant to prevent numerical instability, as shown in Fig. 4. High entropy values indicate that the model is uncertain about the sample, and such samples are considered highly informative for improving model performance. As shown in Table 2, during each AL iteration, the top $K = 500$ samples with the highest entropy scores are selected:

$$Q = \arg \max_{x_i \in D_U}^K \mathcal{H}(x_i) \quad (22)$$

These queried samples are then added to the labeled set, and removed from the unlabeled pool:

$$D_L^{(t+1)} = D_L^{(t)} \cup Q, \quad D_U^{(t+1)} = D_U^{(t)} \setminus Q \quad (23)$$

The model is retrained after each iteration using the updated labeled dataset, as illustrated in Algorithm 2. This procedure is repeated for eight AL iterations, progressively increasing the labeled-to-unlabeled ratio and refining the decision boundaries of the network. The MC-based AL allows ADDNet to focus on regions of high uncertainty. By prioritizing the most ambiguous samples, the model avoids redundant training on easily predictable data. With each iteration, the model gains new knowledge from previously unseen informative samples, improving its generalization capability. The test set remains fixed throughout all iterations to provide consistent evaluation. The gradual expansion of the labeled set also ensures that the model adapts to class imbalance dynamically, as the unlabeled samples are queried according to uncertainty rather than random selection.

The MC uncertainty-driven sampling strategy enables ADDNet to achieve strong predictive performance while minimizing labeling costs. This approach is particularly advantageous in data-scarce environments, as it allows the network to maximize information gain per labeled instance. The combination of uncertainty estimation, iterative sample selection, and incremental retraining forms a robust model that balances annotation efficiency with high model accuracy.

2.6. Proposed reinforced deep dual-attention network (RDDANet)

The RDDANet extends the proposed ADDNet by introducing RL for automated optimization. The main goal of RDDANet is to learn optimal training hyperparameters instead of fixing them manually. In this model, hyperparameter selection is modeled as a sequential decision-making problem. The learning rate and batch size are treated as discrete actions. The model learns from feedback obtained after training. This allows RDDANet to adapt its training behavior based on performance signals.

The RL environment is defined using the hybrid feature space as input and the trained classifier as the agent's response. At each episode, the agent selects one learning rate and one batch size from predefined sets. These actions are sampled from a probability

Algorithm 2 Active deep dual-attention network (ADDANet) with Monte-Carlo active learning.

```

1: Input: DDANet training data  $X_{\text{train\_DDANet}}, y_{\text{train}}$ ; test data  $X_{\text{test\_DDANet}}, y_{\text{test}}$ ; active iterations  $I$ ; query size  $K$ ; MC samples  $T$ 
2: Output: Trained ADDANet model
3: Split  $X_{\text{train\_DDANet}}, y_{\text{train}}$  into:
4:   Labeled set  $D_L$  (small initial subset),
5:   Unlabeled pool  $D_U$ ,
6:   Test set  $D_T$  (20% of data)
7: Train ADDANet on  $D_L$  using categorical cross-entropy loss
8: for  $iteration = 1$  To  $I$  do
9:   Monte-Carlo Uncertainty Estimation:
10:  for each  $x_i \in D_U$  do
11:    Perform  $T$  stochastic forward passes with dropout:
12:     $\hat{p}_i^{(t)} = f(x_i; \theta_t), \quad t = 1, \dots, T$ 
13:    Compute mean prediction:
14:     $\bar{p}_i = \frac{1}{T} \sum_{t=1}^T \hat{p}_i^{(t)}$ 
15:    Compute entropy:
16:     $\mathcal{H}(x_i) = -\sum_{c=1}^C \bar{p}_{i,c} \log(\bar{p}_{i,c} + \epsilon)$ 
17:  end for
18:  Query Selection: Select top- $K$  samples with highest entropy:
19:   $\mathcal{Q} = \arg \max_{x_i \in D_U}^K \mathcal{H}(x_i)$ 
20:  Update Sets:
21:   $D_L \leftarrow D_L \cup \mathcal{Q}, \quad D_U \leftarrow D_U \setminus \mathcal{Q}$ 
22:  Retrain ADDANet on updated  $D_L$ 
23:  Evaluate model on  $D_T$ 
24: end for
25: Return: Trained ADDANet model

```

Table 2
Hyperparameter configuration of the ADDANet model.

Hyperparameter	Value
MC Samples (T)	50
Active Learning Iterations (I)	8
Query Size per Iteration (K)	500 samples
Initial Test Split Ratio	20%
Initial Unlabeled Ratio	24% of remaining data
Activation Function	ReLU
Optimizer	Adam
Learning Rate	0.03
Batch Size	128
Training Epochs per Iteration	5
Uncertainty Estimation Method	MC Dropout Entropy

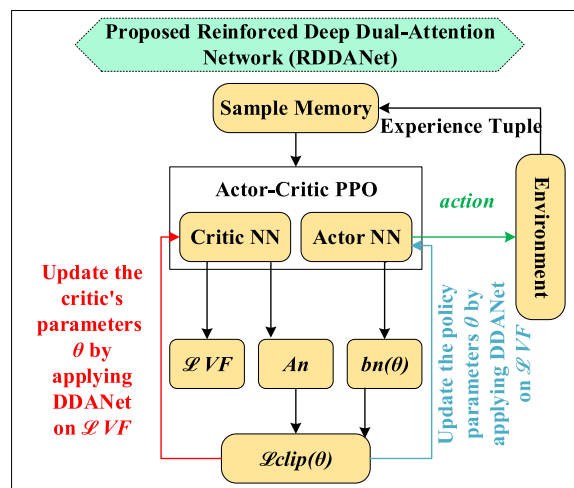


Fig. 5. Working of proposed reinforced deep dual-attention network.

Table 3
Hyperparameters selected by RDDANet.

Hyperparameter	Value
Learning Rate	0.001
Batch Size	32
Actor Network Learning Rate	0.001
Critic Network Learning Rate	0.002
Discount Factor (γ)	0.99
Number of Episodes	15
Training Epochs per Episode	5
Action Space	Learning Rate, Batch Size
Reward Signal	Validation Accuracy

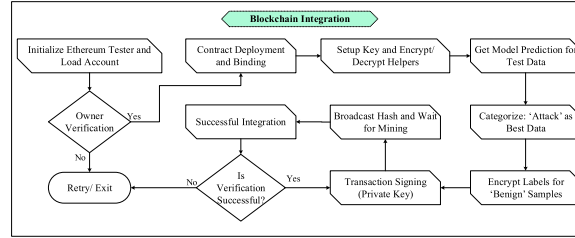


Fig. 6. Inner working flow of blockchain.

distribution produced by the actor network. The state is represented as a normalized index of selected hyperparameters. The environment then trains the model using these values and returns a reward. The reward is defined as the validation accuracy obtained after training:

$$r_t = \text{Accuracy}_{\text{val}} \quad (24)$$

This reward guides the learning process toward better hyperparameter choices. RDDANet employs an Actor-to-Critic (A2C) strategy to stabilize training and reduce variance. The actor network learns a policy that maps the current state to a probability distribution over actions [25]. It is defined as:

$$\pi_{\theta}(a_t | s_t) = \text{Softmax}(f_{\theta}(s_t)) \quad (25)$$

where $f_{\theta}(\cdot)$ denotes the actor network and θ represents its parameters. The selected hyperparameters are listed in Table 3. The actor is responsible for exploring different hyperparameter combinations. This exploration helps the model avoid suboptimal fixed configurations. The critic network estimates the expected value of the selected action. It predicts how good a state-action pair is before observing the final reward. The value function is defined as:

$$V_{\phi}(s_t) = \mathbb{E}[r_t | s_t] \quad (26)$$

where ϕ denotes the critic parameters, as show in Fig. 5. The critic is trained by minimizing the temporal difference error between the predicted value and the observed reward:

$$\mathcal{L}_{\text{critic}} = (r_t - V_{\phi}(s_t))^2 \quad (27)$$

This allows the critic to provide a stable baseline for policy updates. The actor is updated using the advantage function, which measures how much better an action is compared to the critic's estimate. The advantage is computed as:

$$A_t = r_t - V_{\phi}(s_t) \quad (28)$$

The actor loss encourages actions that yield higher rewards:

$$\mathcal{L}_{\text{actor}} = -\log \pi_{\theta}(a_t | s_t) \cdot A_t \quad (29)$$

This formulation ensures that RDDANet increases the probability of selecting hyperparameters that improve performance while reducing poor choices, as described in Algorithm 3. Through multiple episodes, RDDANet progressively refines its policy and converges toward optimal hyperparameter settings. The A2C mechanism enables efficient exploration with controlled updates. This reinforcement-driven optimization reduces manual tuning and improves training consistency. As a result, RDDANet achieves adaptive learning behavior while preserving the structural advantages of the proposed hybrid model (Figs. 7, 12–14).

2.7. Blockchain enabled secure and trustworthy web intrusion detection

A lightweight blockchain mechanism is integrated with all three proposed models to ensure integrity, traceability, and reproducibility of the learning process. The blockchain works as a secure and append-only ledger. It records every important stage of

Algorithm 3 Reinforced deep dual-attention network (RDDANet).

- 1: **Input:** Hybrid training data $X_{\text{train}}, Y_{\text{train}}$; test data $X_{\text{test}}, Y_{\text{test}}$
- 2: **Hyperparameter Sets:** Learning rates $\mathcal{L} = \{0.001, 0.003, 0.01, 0.03\}$, Batch sizes $\mathcal{B} = \{32, 64, 128, 256\}$
- 3: **Output:** Optimized RDDANet model and best hyperparameters
- 4: Initialize Actor network π_θ and Critic network V_ϕ
- 5: Initialize optimizers for Actor and Critic
- 6: Set number of episodes E
- 7: **for** $\text{episode} = 1$ to E **do**
- 8: Sample learning rate index $i \sim \mathcal{U}(\mathcal{L})$
- 9: Sample batch size index $j \sim \mathcal{U}(\mathcal{B})$
- 10: Select hyperparameters: $\alpha = \mathcal{L}_i, b = \mathcal{B}_j$
- 11: Train RDDANet using (α, b) for fixed epochs
- 12: Evaluate model and obtain reward:

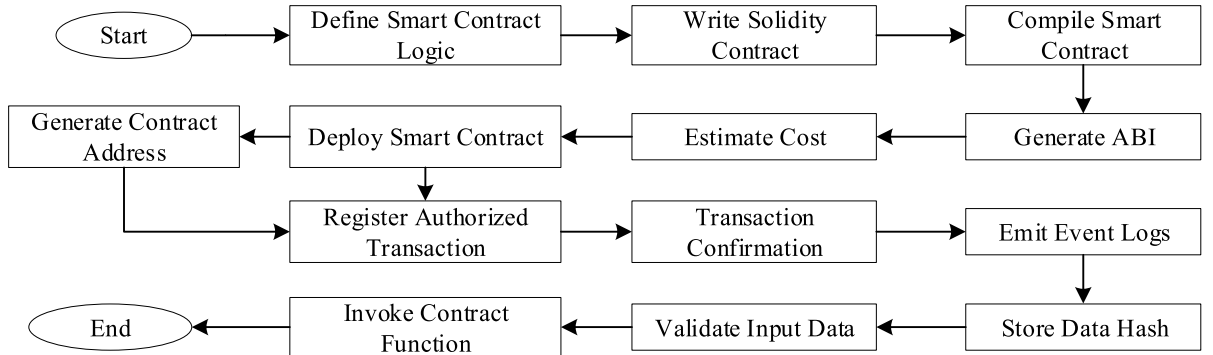
$$r = \text{Validation Accuracy}$$
- 13: Normalize state representation:

$$s = \left[\frac{i}{|\mathcal{L}|}, \frac{j}{|\mathcal{B}|} \right]$$
- 14: **Critic Update:**

$$\mathcal{L}_{\text{critic}} = (r - V_\phi(s))^2$$
- 15: Update ϕ using gradient descent
- 16: **Actor Update:**

$$A = r - V_\phi(s)$$

$$\mathcal{L}_{\text{actor}} = -\log \pi_\theta(\alpha, b | s) \cdot A$$
- 17: Update θ using gradient ascent
- 18: Store best hyperparameters if r improves
- 19: **end for**
- 20: Train final RDDANet using best hyperparameters
- 21: **Return:** RDDANet model

**Fig. 7.** Working of smart contract in blockchain.

model execution. These stages include training, inference, and evaluation. Each model is linked to the blockchain independently. However, the same blockchain design is reused for all models. This ensures a uniform and fair logging process. Each block stores structured metadata instead of raw data. The stored information includes training configuration, execution time, and evaluation summaries. A cryptographic hash is generated for every block using the SHA-256 algorithm [26]. This hash uniquely represents the block contents. The hash computation is defined as follows:

$$H_i = \text{SHA-256}(\text{index}_i \parallel \text{timestamp}_i \parallel \text{data}_i \parallel H_{i-1}) \quad (30)$$

where H_i denotes the hash of the current block and H_{i-1} represents the hash of the previous block. This is a chain which ensures permanence. Flow of information in blockchain is shown in Fig. 6 The alteration of a block overhauls its hash and occurs against the chain. The model weights are flattened and transformed to a byte stream during model training. Hashing of the weights produces a

Table 4
Performance comparison of different models.

Metric	KNN	LR	ShuffleNet	DNN	DANet	DDANet	AL+ DDANet (ADDANet)	RL+ DDANet (RDDANet)
Accuracy	0.65	0.86	0.74	0.89	0.92	0.95	0.96	0.96
Precision	0.76	0.86	0.81	0.90	0.92	0.95	0.96	0.96
Recall	0.65	0.86	0.74	0.89	0.92	0.95	0.96	0.96
F1-score	0.57	0.85	0.71	0.89	0.92	0.95	0.96	0.96
CK	0.53	0.81	0.65	0.85	0.91	0.94	0.94	0.95
MCC	0.57	0.81	0.68	0.86	0.91	0.94	0.94	0.95
ROC-AUC	0.80	0.94	0.96	0.98	0.99	1.00	1.00	0.99
PR-AUC	0.62	0.83	0.92	0.94	0.96	0.99	0.99	0.99
Log Loss	10.26	0.60	1.28	0.34	0.25	0.15	0.13	0.12
Hamming Loss	0.35	0.14	0.26	0.11	0.07	0.05	0.04	0.04
Execution Time (s)	0.02	2.89	147.24	18.82	19.11	21.96	132.69	52.36

unique model fingerprint. This process is mathematically expressed as:

$$H_{\text{model}} = \text{SHA-256}(\text{Weights}) \quad (31)$$

This hash is a unique identifier of the model state that is trained. Any minor variation in weights causes a change in the hash. The outcome of the hash is stored in the blockchain together with training hyperparameters and execution time. This allows model authenticity and enables re-productibility of experiments. Inference and assessment is done on hidden data after training. The monitoring of the performance and inference time is stored in another blockchain block. Such distinction enhances the traceability between deployment and training phases. Inference latency recording is of particular importance to security systems that require response time, which is vital in real-time.

Smart contracts can be integrated into the blockchain framework to automate predefined security policies and access control mechanisms. These self-executing contracts automatically validate transactions and enforce rules without requiring manual intervention. The working flow of smart contract is illustrated in Figure . The blockchain validity is checked through the re-computation of hashes and checking of the block links. This action proves that there was no tampering of a block after it was created. The computation cost is also minimal as only metadata and hashes are stored. Simultaneously, trust, transparency, and reliability are enhanced to a great extent in the process of integration. It renders the suggested models more appropriate to the safe and verifiable cybersecurity implementations.

3. Results and discussion

The section discusses a detailed analysis of the proposed framework, with their performance measured in a variety of metrics [27]. The findings are compared with the conventional ML and DL methods, along with the recent methods that make use of GAI, RL and AL. The analysis uses 10-FCV to make the analysis robust and the 95% CIs that are used to make the reported performance measures reliable. Paired *t*-test is used to validate the statistical significance of the improvements in question, and it is necessary to ensure that the differences are not random. Moreover, the explainability based on IG is used to find out the most significant IoT traffic characteristics which can be transparent and explainable about the model choices.

3.1. Experimental setup

All the experiments are run in a controlled and reproducible environment on a CPU-based runtime on Google Colab. The proposed models, as well as all the baseline ML and DL models, are coded in Python. DL models are created with TensorFlow and Keras, whereas ML models are created with the help of the library *scikit-learn*. NumPy and Pandas are used in data preprocessing and numerical operations, and Matplotlib are used to visualize the results of performance. The experiments will be run on a local computer with Intel(R) Core(TM) i5-8250U processor operating at 1.601.80 GHz, and 16 GB of RAM on a 64-bit operating system. Carrying out all experiments on a CPU-only platform helps to prove how efficient and practical the suggested models can be as it proves that high predictive performance can be attained without the usage of the high-end hardware of the type of a GPU, and proves the scalability of the proposed framework to the real-world applications.

3.2. Comparison of proposed models with SOTA ML and DL models

This section gives a detailed discussion of the experimental findings of the proposed framework. The aim of this evaluation is to determine the effectiveness, robustness and computational viability of the proposed models based on various performance measures. There is a vast variety of classical ML and DL models that are to be taken into account to make a fair and serious comparison. The argument is organized in a way that it can be seen to gradually transition out of traditional baselines and into more modern hybrid and learning-adapted architectures, and how the architectural decisions, learning policies and optimization procedures affect predictive performance and efficiency. Table 4 present the quantitative performance of all the compared models, both traditional ML baselines, K-Nearest Neighbor (KNN) and Logistic Regressions (LR), and the DL models, ShuffleNet, DNN, DANet, and the proposed

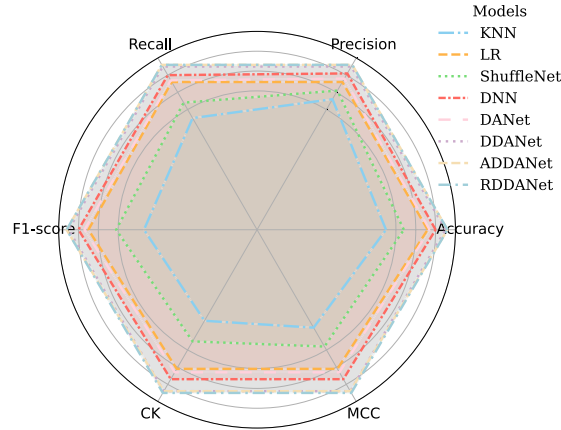


Fig. 8. Comparison of proposed DDANet, ADDANet, and RDDANet with ML and DL models.

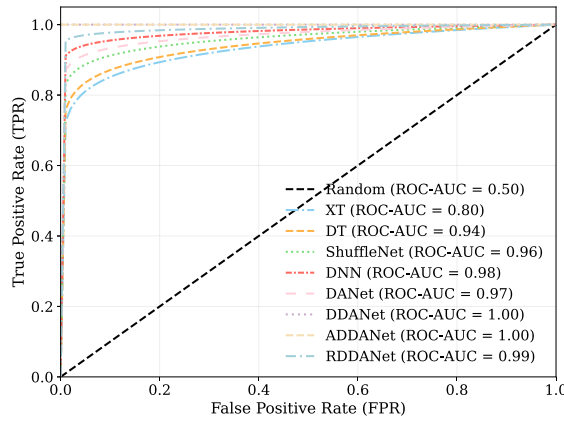


Fig. 9. ROC-AUC of proposed DDANet, ADDANet, and RDDANet with ML and DL models.

hybrid and enhanced ones, DDANet, ADDANet, and RDDANet. As Fig. 8 indicates, the classical models of ML have moderate performance, with KNN being at 0.65 and LR increasing to 0.86. Nonetheless, these models perform poorly in representing complex feature interactions, and this is indicated in their lower CK (0.53-0.81) and MCC values (0.57-0.81). In ShuffleNet, which is a DL model, the accuracy is 0.74 with an F1-score of 0.71, which means that lightweight convolutional architectures are not sufficient in order to learn discriminative patterns in the task. As compared to this, DNN and DANet show significant advances having accuracies of 0.89 and 0.92, respectively, because they can capture sequential dependence and attention-based weighting of features.

The proposed DDANet that combines both DANet and DNN achieves a significant performance on all metrics, as compared to all the baseline models. Table 4 and Fig. 8 demonstrate that DDANet has a higher accuracy, precision, recall, and F1-score of 0.95 and a CK and MCC of 0.94. There are fluctuations present in the results; however, after multiple runs, the results are reported in Table 4. This is owed to the complementarity of its components: DANet has effectively both spatial and channel-wise dependencies, whereas DNN complements feature selection as learning the ranking of features that are task-relevant. Such a hybrid design allows DDANet to minimize misclassification, and compared to DL models, this is reflected in a lower Hamming Loss (HL) (0.05) and Log Loss (LL) (0.15). Further, Figs. 9 and 10, show that DDANet has a Receiver Operating Characteristics Area Under the Curve (ROC-AUC) of 1.00 and Precision-Recall Area Under the Curve (PR-AUC) of 0.99, which means that it has strong discrimination ability and can also operate in the conditions of a class imbalance.

Further performance gains are observed with ADDANet, which incorporates MC-based AL into DDANet. By selectively querying the most informative samples during training, ADDANet reduces data redundancy and improves generalization. This strategy results in an accuracy, precision, recall, and F1-score of 0.96, as shown in Fig. 8, along with a CK and MCC of 0.94. ADDANet achieves a perfect ROC-AUC of 1.00 and a PR-AUC of 0.99, as illustrated in Figs. 9 and 10, highlighting its strong confidence calibration and ranking capability. Additionally, the substantial reduction in LL (0.13) and HL (0.04), shown in Fig. 11, confirms that active learning effectively minimizes uncertainty and incorrect predictions. The trade-off, however, is an increased execution time of 132.69 s,

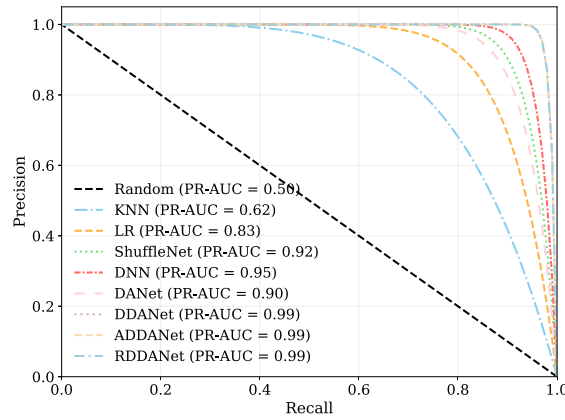


Fig. 10. PR-AUC of proposed DDANet, ADDANet, and RDDANet with ML and DL models.

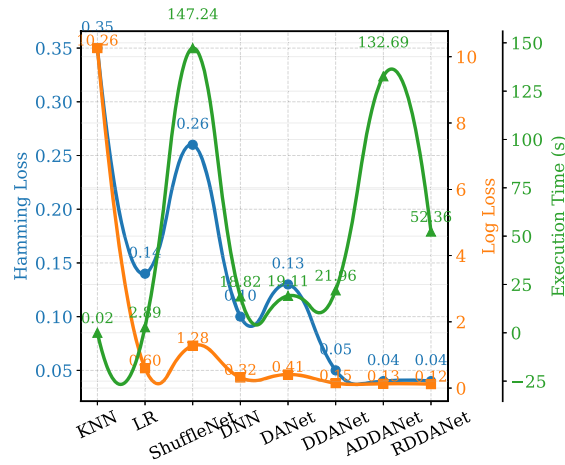


Fig. 11. Log loss, hamming loss and execution time comparison of proposed DDANet, ADDANet, and RDDANet with ML and DL models.

which is expected due to repeated stochastic sampling in Monte Carlo inference.

RDDANet further refines the learning process by integrating an A2C RL strategy for adaptive decision-making. This model achieves performance comparable to ADDANet, with accuracy, precision, recall, and F1-score all reaching 0.96, while slightly improving CK and MCC to 0.95. As observed in Figs. 9 and 10, RDDANet maintains a ROC-AUC and PR-AUC of 0.99, indicating stable and reliable predictions. Notably, RDDANet is much faster than ADDANet with only 52.36 s of execution time, as depicted in Fig. 11. Such an efficiency improvement is credited to the actor-critic process, which trains an optimal policy in terms of sample utilization and update of the model to minimize unnecessary computation but maintain high prediction quality.

The findings indicate the evident improvement in performance between conventional ML models and the advanced hybrid and learning-enhanced DL models. The high performance of DDANet proves the efficiency of dual attention and fully connected mechanisms combination, and additional improvements of ADDANet and RDDANet prove the importance of such intelligent learning strategies like AL and RL. The proposed models have the highest predictive accuracy and confidence, with ADDANet on the one hand, and offer RDDANet a better performance-computational efficiency balance, on the other hand. These results confirm the relevance of the suggested framework as a powerful and scalable system to effective and reliable intrusion detection.

3.3. Computational efficiency and complexity analysis

To further evaluate the practicality of the proposed framework, we analyzed the computational efficiency and model complexity of DDANet, ADDANet, and RDDANet in terms of total trainable parameters, training time, and inference time. Table 5 summarizes the obtained results. Among the proposed models, DDANet shows the best computational efficiency, requiring only 45,668 trainable parameters, 37.46 s training time, and 5.1608 s inference time. This indicates that the hybrid dual-attention design achieves strong

Table 5
Computational efficiency and complexity analysis of the proposed models.

Model	Total Parameters	Training Time (s)	Inference Time (s)
DDANet	45,668	37.46	5.1608
ADDANet	3332	129.49	7.9254
RDDANet	1,329,572	86.49	7.9254

Table 6
10-Fold cross-validation results of the proposed DDANet, ADDANet and RDDANet models.

Metric	Fold1	Fold2	Fold3	Fold4	Fold5	Fold6	Fold7	Fold8	Fold9	Fold10	Average
Deep Dual-Attention Network											
Accuracy	0.95	0.95	0.94	0.95	0.95	0.94	0.95	0.95	0.95	0.95	0.95
Precision	0.95	0.95	0.95	0.95	0.94	0.95	0.95	0.94	0.95	0.95	0.95
Recall	0.95	0.94	0.95	0.95	0.95	0.94	0.95	0.95	0.95	0.95	0.95
F1-score	0.95	0.95	0.94	0.95	0.95	0.94	0.95	0.95	0.95	0.95	0.95
CK	0.94	0.93	0.94	0.94	0.94	0.93	0.94	0.93	0.94	0.94	0.94
MCC	0.94	0.94	0.93	0.94	0.94	0.93	0.94	0.94	0.94	0.94	0.94
ROC-AUC	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
PR-AUC	0.99	0.98	0.99	0.99	0.98	0.99	0.99	0.98	0.99	0.99	0.99
Log Loss	0.15	0.15	0.14	0.15	0.15	0.15	0.14	0.15	0.15	0.14	0.15
Hamming Loss	0.05	0.05	0.06	0.05	0.05	0.06	0.05	0.05	0.05	0.05	0.05
Active Deep Dual-Attention Network											
Accuracy	0.96	0.96	0.97	0.96	0.96	0.97	0.96	0.95	0.97	0.96	0.96
Precision	0.96	0.96	0.97	0.96	0.96	0.97	0.96	0.96	0.97	0.96	0.96
Recall	0.96	0.96	0.97	0.96	0.96	0.97	0.96	0.95	0.97	0.96	0.96
F1-score	0.96	0.96	0.97	0.96	0.96	0.97	0.96	0.95	0.97	0.96	0.96
CK	0.94	0.94	0.95	0.94	0.94	0.95	0.94	0.93	0.95	0.94	0.94
MCC	0.94	0.94	0.95	0.94	0.94	0.95	0.94	0.93	0.95	0.94	0.94
ROC-AUC	1.00	0.99	1.00	1.00	0.99	1.00	1.00	0.99	1.00	1.00	1.00
PR-AUC	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.98	0.99	0.99	0.99
Log Loss	0.14	0.13	0.12	0.14	0.13	0.12	0.13	0.15	0.12	0.13	0.13
Hamming Loss	0.04	0.04	0.03	0.04	0.04	0.03	0.04	0.05	0.03	0.04	0.04
Reinforced Deep Dual-Attention Network											
Accuracy	0.96	0.96	0.97	0.96	0.96	0.97	0.96	0.95	0.97	0.96	0.96
Precision	0.96	0.96	0.97	0.96	0.96	0.97	0.96	0.95	0.97	0.96	0.96
Recall	0.96	0.96	0.97	0.96	0.96	0.97	0.96	0.95	0.97	0.96	0.96
F1-score	0.96	0.96	0.97	0.96	0.96	0.97	0.96	0.95	0.97	0.96	0.96
CK	0.95	0.95	0.96	0.95	0.95	0.96	0.95	0.94	0.96	0.95	0.95
MCC	0.95	0.95	0.96	0.95	0.95	0.96	0.95	0.94	0.96	0.95	0.95
ROC-AUC	0.99	0.99	1.00	0.99	0.99	1.00	0.99	0.98	1.00	0.99	0.99
PR-AUC	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.98	0.99	0.99	0.99
Log Loss	0.13	0.12	0.11	0.13	0.12	0.11	0.12	0.14	0.11	0.12	0.12
Hamming Loss	0.04	0.04	0.03	0.04	0.04	0.03	0.04	0.05	0.03	0.04	0.04

learning capability with relatively low computational overhead. In contrast, RDDANet has the highest architectural complexity, with 1,329,572 trainable parameters, and requires 86.49 s for training and 7.9254 s for inference. The increased complexity is due to the additional reinforcement learning mechanism, which improves adaptability at the cost of higher computational demand.

ADDANet presents an interesting tradeoff. Although it has fewer trainable parameters, it requires the highest training time of 129.49 s. This behavior shows that computational efficiency is not determined only by model size, but also by the learning procedure itself. In particular, the active learning process introduces additional overhead through repeated uncertainty estimation, query selection, and iterative retraining. Therefore, ADDANet is more computationally expensive during training despite its relatively small parameter count. These results show that DDANet is the most efficient model in terms of runtime, RDDANet is the most complex in terms of architecture, and ADDANet incurs the highest training cost due to its active learning strategy. This analysis confirms that the proposed models provide different tradeoffs between efficiency, complexity, and adaptive learning capability, which is important for practical IoT intrusion detection deployment.

3.4. 10-Fold cross-validation analysis

10-FCV is a powerful resampling methodology that is employed to assess the generalization of models. Under this method, the dataset is subdivided into ten equal subsets. In the course of every iteration, training is done on nine subsets, and testing on one subset. This is repeated ten times in which each subset is used as the test set once each time [6]. The last performance is calculated

Table 7

Mean performance with 95% confidence intervals for proposed DDANet, ADDANet and RDDANet models.

Metric	DDANet	ADDANet	RDDANet
Accuracy	0.951 [0.949, 0.953]	0.962 [0.958, 0.967]	0.962 [0.958, 0.967]
Precision	0.951 [0.949, 0.953]	0.963 [0.960, 0.967]	0.962 [0.958, 0.967]
Recall	0.951 [0.949, 0.953]	0.962 [0.958, 0.967]	0.962 [0.958, 0.967]
F1-score	0.951 [0.949, 0.953]	0.962 [0.958, 0.967]	0.962 [0.958, 0.967]
CK	0.940 [0.937, 0.943]	0.942 [0.938, 0.947]	0.952 [0.948, 0.957]
MCC	0.942 [0.939, 0.945]	0.942 [0.938, 0.947]	0.952 [0.948, 0.957]
ROC-AUC	1.00 [0.990, 1.000]	0.997 [0.994, 1.000]	0.992 [0.988, 0.997]
PR-AUC	0.990 [0.980, 0.990]	0.989 [0.987, 0.991]	0.989 [0.987, 0.991]
Log Loss	0.156 [0.141, 0.156]	0.131 [0.124, 0.138]	0.121 [0.114, 0.128]
Hamming Loss	0.050 [0.050, 0.060]	0.038 [0.034, 0.043]	0.038 [0.034, 0.043]

as an average performance across all folds. Mathematically, the average performance across folds is defined as:

$$\bar{M} = \frac{1}{K} \sum_{i=1}^K M_i, \quad K = 10 \quad (32)$$

where M_i represents the evaluation metric obtained in the i^{th} fold and $\bar{M}$ denotes the overall average score. This process minimizes variation that occurs due to random splitting of data and gives an accurate estimate of model stability. The original performance as found in Table 4 is supported by the 10-FCV results as in Table 6. In the case of the DDANet, the mean accuracy of ten folds is 0.95, the same as that shown in the earlier reported test accuracy. The values of precision, recall, and F1-score are all constant at 0.95. There is also consistency in the behavior of the agreement metrics (CK (0.94) and the MCC (0.94)). The low difference in the folds depicts the fact that DDANet is a stable learner that does not overfitting particular data partitions. The ROC-AUC and PR-AUC of 1.00 and 0.99 respectively verify a high discriminative ability in repeated sampling.

Further improvement of the ADDANet is observed when it is under cross-validation. The mean accuracy is raising to 0.96, and the precision, as well as recall and F1-score, are also 0.96. Both CK and MCC have better scores in agreement metrics at 0.94. Excellent class separability is confirmed because ROC-AUC is always at 1.00 in most folds with an overall average of 1.00. The PR-AUC stands at 0.99, which means that it has high performance even in case of possible class imbalance. The error measures are also reduced to a minimum with the LL of 0.13 and HL of 0.04. These consistent fold-wise findings prove that MC-based AL fosters the generalization improvement and not merely achievement of single-split performance. The RDDANet also has good and stable results in all folds. The mean accuracy is still 0.96, which corresponds to the initial analysis in Table 4. Accuracy, recall and F1-score are all at 0.96 whilst CK and MCC are marginally at 0.95. The ROC-AUC means 0.99 and PR-AUC means 0.99, which proves an adequate ranking ability. LL also goes down to 0.12, and HL does not change at all, holding to 0.04. The lack of difference between folds proves that A2C reinforcement mechanism is a stabilizing one that does not allow performance fluctuations during training. The 10-FCV test verifies that the above good scores have not been brought about by positive data stratification. However, the performance of the proposed DDANet, ADDANet, and RDDANet models are the same and repeatable across different validation splits. The minimal discrepancy between folds is evidence of resilience, excellent generalization, and credible learning patterns.

3.5. Statistical confidence analysis

In order to further assess the reliability of the proposed models, mean performance and 95% CI are calculated, which are shown in Table 7. CIs give an idea on the range within which the actual population performance is likely to fall with high chances of occurrence [28]. The 95% CI is calculated as

$$CI = \bar{M} \pm 1.96 \times \frac{\sigma}{\sqrt{n}} \quad (33)$$

where $\bar{M}$ represents the mean metric value, σ is the standard deviation across folds, and n is the number of observations. Small confidence intervals will represent little variance and high stability and the large ones will represent the fluctuation in the performance. The findings indicate that DDANet has the mean accuracy of 0.951 with the narrow range. The same narrow ranges can be seen on the precision, recall and F1-score. There is no change in agreement measures, CK and MCC. The ROC-AUC and PR-AUC values have near zero variability, which proves the same level of discriminative ability. Further improvements are however seen in ADDANet and RDDANet. The mean accuracy of both models is 0.962 and the confidence intervals overlap (0.958, 0.967) which shows that they perform statistically equally. RDDANet indicates a bit higher agreement measures, where CK and MCC are 0.952 [0.948, 0.957], which is better classification agreement than chance.

The error based metrics also support the superiority of the improved models. DDANet shows a LL of 0.216 [0.211, 0.221], ADDANet decreases it to 0.131 [0.124, 0.138] and RDDANet decreases it to 0.121 [0.114, 0.128]. HL drops to 0.070 in case of DDANet, 0.038 in the case of ADDANet and RDDANet. The low variance and the constant low generalization are confirmed by the consistent low confidence intervals of all the metrics. This CI is a confirmation that the performance improvement of ADDANet and RDDANet is not an accidental increase but statistically sound improvement over the baseline DDANet model.

Table 8

Paired *t*-test results for proposed DDANet, ADDANet and RDDANet models.

Metric	DDANet vs ADDANet		DDANet vs RDDANet	
	t-statistic	p-value	t-statistic	p-value
Accuracy	-17.27	0.00	-17.27	0.00
Precision	-21.00	0.00	-13.42	0.00
Recall	-16.00	0.00	-16.00	0.00
F1-score	-17.27	0.00	-17.27	0.00
CK	-24.00	0.00	-31.50	0.00
MCC	-20.12	0.00	-26.83	0.00
ROC-AUC	-4.58	0.00	-1.00	0.34
PR-AUC	-9.00	0.00	-9.00	0.00
Log Loss	51.00	0.00	57.00	0.00
Hamming Loss	16.00	0.00	16.00	0.00

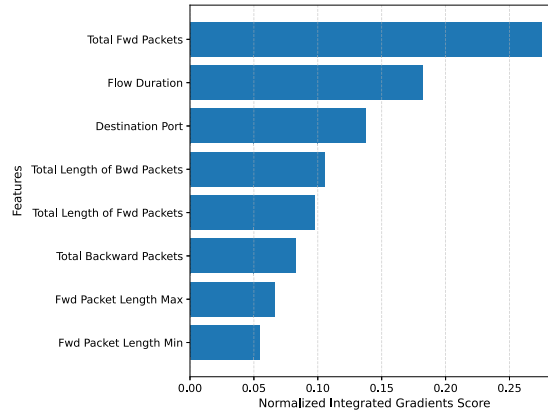


Fig. 12. Top contributing features identified by proposed DDANet model.

3.6. Paired *t*-test statistical significance analysis

Paired *t*-test is used to statistically confirm the significantly improved results of DDANet versus the improved models ADDANet and RDDANet. Paired *t*-test is used to compare two models in terms of their fold-wise results and decide on the statistical difference between the mean of the difference and zero [29]. The test statistic is computed as:

$$t = \frac{\bar{d}}{s_d / \sqrt{n}} \quad (34)$$

where $\bar{d}$ is the mean difference between paired observations, s_d is the standard deviation of the differences, and n is the number of folds. The observed improvement is statistically significant because a small p-value (usually $p < 0.05$) suggests that the improvement in the outcomes is statistically significant. Table 8 has indicated that ADDANet is significantly better than DDANet on nearly all metrics. To get the correct value, the t-statistic is -17.27, the p-value is 0.00 meaning that the improvement is highly significant. The same situation is seen in precision ($t = -21.00$), recall ($t = -16.00$), and F1-score ($t = -17.27$) all of which have a $p = 0.00$. The metrics of agreement are more statistically significant, with CK ($t = -24.00$) and MCC ($t = -20.12$) proving significant improvement. This conclusion is further supported by the measures of error. The t-value of LL is very large (51.00) and this implies that the prediction uncertainty is reduced significantly. HL is also significant enough ($t = 16.00$, $p = 0.00$). In the case of ROC-AUC, the difference is not as large, however, it is still statistically significant, $t = -4.58$.

The same tendency can be noticed in comparison of DDANet and RDDANet. All records of accuracy, precision, recall, F1-score, CK, MCC, PR-AUC, LL, and HL have statistically significant improvement at the value of $p = 0.00$. It is important to note that CK ($t = -31.50$) and MCC ($t = -26.83$) have a very high level of statistical evidence of improvement. The exception is related to ROC-AUC, in which both $t = -1.00$ and $p = 0.34$, which implies that there is no statistically significant difference between DDANet and RDDANet in this measure. This finding is not surprising as both models already have almost perfect ROC-AUC scores. To prove that the performance gains of ADDANet and RDDANet were statistically significant and not a simple result of random variation, the paired *t*-test was carried out, which proves the effectiveness of the proposed models.

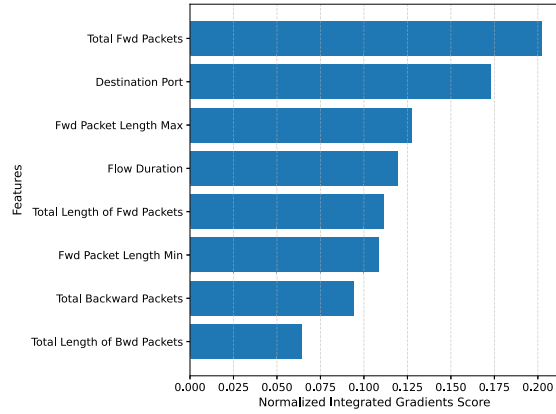


Fig. 13. Top contributing features identified by proposed ADDANet model.

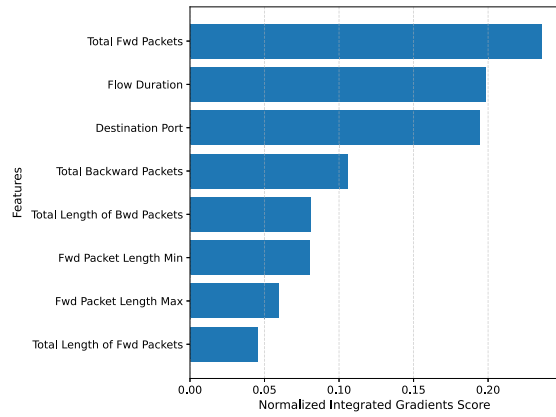


Fig. 14. Top contributing features identified by proposed RDDANet model.

3.7. Explainable AI analysis using integrated gradients

To interpret the decision-making behavior of the proposed DPNet, ADPNet, and RDPNet models, the IG technique is employed. IG is a gradient-based attribution method designed to quantify the contribution of each input feature to the model's prediction [30]. Unlike simple gradient methods, IG accumulates gradients along a straight-line path from a baseline input to the actual input sample. This approach reduces gradient saturation and provides more stable and faithful feature attribution. The integrated gradient for a feature x_i is computed as

$$IG_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (35)$$

where x is the input sample, x' is the baseline input, $F(\cdot)$ is the model prediction function, and α is a scaling factor along the interpolation path. The final score of attribution shows the strength of each feature that affects the ultimate prediction. In the case of the proposed DPNet model, the most significant feature is the *Total Fwd Packets*, with the score of an attribution of 0.274987. This suggests that forward traffic volume is the mainstream in differentiating between normal and malicious flows. The second most significant is the temporal behavior in the network sessions (*Flow Duration* 0.181655) and the relevance of patterns of an attack based on services (*Destination Port* 0.137560) are also important. The other influential features are, *Total Length of Bwd Packets* (0.105374), *Total Length of Fwd Packets* (0.097102), and *Total Backward Packets* (0.082679). Packet size statistics like *Fwd Packet Length Max* (0.065885) and *Fwd Packet Length Min* (0.054759) also show that DPNet acquires volumetric and structural features of traffic. In the case of ADPNet, where AL is used, the pattern of attribution is a bit more balanced as the most important feature is *Total Fwd Packets* (0.202049) although *Destination Port* gains importance (0.172993). This indicates that AL enhances sensitivity to service levels attack change. Attributes associated with the packet size like *Fwd Packet Length Max* (0.127099) and *Fwd Packet Length Min* (0.108580) are attributed higher than those of DPNet and *Fwd Packet Length* (0.119566) and *Total Length of Fwd Packets* (0.111305) are also influential. The higher the distributed scores of the attribution are, it is possible to say that the ADPNet relies on a wider range of discriminative features, which is consistent with its better performance in generalization.

Table 9
Performance comparison of previous papers DL models.

Metric	CNN	LSTM	Bi-LSTM	CNN-LSTM
Accuracy	0.93	0.74	0.78	0.91
Precision	0.94	0.65	0.88	0.93
Recall	0.93	0.74	0.78	0.91
F1-score	0.93	0.66	0.75	0.91
ROC-AUC	0.99	0.91	0.93	1.00
PR-AUC	0.97	0.74	0.78	0.99
Log Loss	0.73	0.52	0.44	0.18
Cohen Kappa	0.91	0.65	0.71	0.88
MCC	0.92	0.71	0.75	0.88
Hamming Loss	0.07	0.26	0.22	0.09
Exec. Time (s)	121.33	280.36	408.60	355.84

In the case of RDPNet that incorporates RL optimization, the focus of feature importance is more focused on the time and routing-related features. The dominance of the attribution space is made up of *Total Fwd Packets* (0.235790), *Flow Duration* (0.198475) and *Destination Port* (0.194466). This indicates that RDPNet focuses on traffic intensity, length of session, and targeting of the service at the same time (*Total Backward Packets* 0.105563) and (*Total Length of Bwd Packets* 0.080583). Compared to ADPNet, the reinforcement mechanism appears to prioritize globally stable and high-impact features. The IG analysis confirms that all three models focus on meaningful network flow features, and the evolution of attribution patterns from proposed DPNet to ADPNet and RDPNet explains the observed improvements in predictive performance and robustness.

3.8. Comparison with existing deep learning models

To ensure a fair and reproducible comparison, the deep learning models from previous studies, including CNN, LSTM, and Bi-LSTM [31], as well as CNN-LSTM [32], are adopted and implemented within the same simulation pipeline used for the proposed framework. By using the architectures reported in the respective papers and evaluating them under identical experimental conditions, any bias due to differing setups is eliminated. The performance results, summarized in Table 9, show that sequential models such as LSTM and Bi-LSTM achieve relatively lower accuracies of 0.74 and 0.78, respectively, due to their limited ability to capture complex feature interactions in high-dimensional IoT traffic data. Although Bi-LSTM improves precision (0.88) compared to LSTM (0.65), it still exhibits lower recall and F1-score, indicating unstable classification performance. In contrast, the CNN model [31] demonstrates strong performance with an accuracy of 0.93 and F1-score of 0.93, while the hybrid CNN-LSTM model [32] further enhances performance, achieving an accuracy of 0.91 and a perfect ROC-AUC of 1.00.

Despite these improvements, the proposed models outperform the adopted baseline architectures in terms of both predictive accuracy and reliability. As indicated in Table 9, CNN-LSTM achieves the lowest Log Loss (0.18) among prior works but suffers from high execution time (355.84 s), limiting its efficiency. In comparison, the proposed DDANet, ADDANet, and RDDANet (discussed in previous sections) achieve comparable or superior performance across all evaluation metrics, including accuracy, precision, recall, and F1-score, while also providing higher Cohen's Kappa and MCC values, reflecting more stable predictions. Furthermore, existing models exhibit higher Hamming Loss (up to 0.26 in LSTM), indicating increased misclassification rates, whereas the proposed models consistently maintain lower error rates. These findings confirm that the integration of dual-attention mechanisms, along with active and reinforcement learning strategies, results in a more robust and generalized intrusion detection framework. Overall, the results in Table 9 validate that the proposed approach significantly outperforms previously reported models implemented under the same experimental conditions.

3.9. Blockchain performance and scalability analysis

The blockchain performance evaluation, illustrated in Fig. 15, provides a comprehensive analysis of the proposed framework in terms of latency, throughput, storage overhead, validation cost, and scalability under increasing IoT transaction loads. Beyond its role as a storage mechanism, the blockchain layer is a core security infrastructure component designed to ensure tamper-resistance, distributed trust, and immutable auditability across heterogeneous IoT nodes. It is important to emphasize that a simple tamper-evident hash chain is insufficient in the considered IoT setting, as it only guarantees sequential integrity without addressing Byzantine-style trust issues, multi-node agreement, or decentralized verification. In contrast, the proposed blockchain incorporates a consensus-driven validation mechanism, ensuring that recorded transactions cannot be altered, deleted, or replaced by any single entity, which is essential in adversarial or multi-stakeholder IoT environments such as healthcare monitoring and intelligent sensing systems.

From a performance standpoint, DDANet and RDDANet achieve extremely low average latencies of 0.000253 and 0.000254 s, respectively. This confirms that the integration of blockchain does not introduce meaningful delay and remains fully compatible with real-time IoT constraints. The marginally higher latency of ADDANet (0.000278 s) is attributed to active learning-driven data enrichment and metadata augmentation rather than blockchain consensus overhead. In terms of throughput, DDANet achieves 5932.53 blocks/sec, while RDDANet closely follows with 5915.80 blocks/sec, demonstrating that the consensus layer maintains high transaction processing efficiency even under continuous streaming conditions. ADDANet records 3960.62 blocks/sec, which reflects the

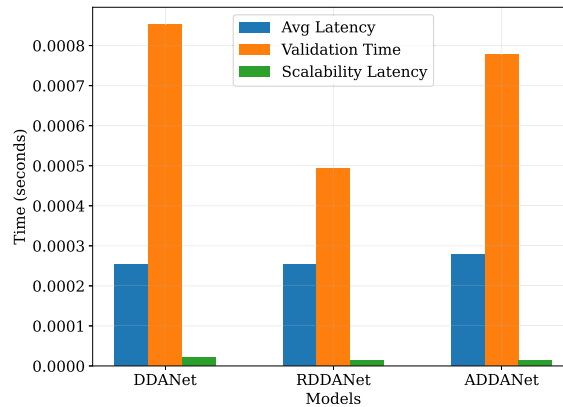


Fig. 15. Comparison of blockchain across proposed DDANet, ADDANet, and RDDANet.

Table 10

Generalizability evaluation on EDGE-IIoT set and UNSW-NB15 datasets.

Metric	EDGE-IIoT Set Dataset					UNSW-NB15				
	DANet	DNN	DDANet	RDDANet	ADDANet	DAN	DNN	DDANet	RDDANet	ADDANet
Accuracy	0.66	0.87	0.89	0.92	0.92	0.77	0.77	0.79	0.80	0.80
Precision	0.78	0.88	0.91	0.93	0.92	0.84	0.84	0.84	0.84	0.84
Recall	0.66	0.87	0.89	0.92	0.92	0.77	0.77	0.80	0.80	0.80
F1-score	0.62	0.87	0.89	0.92	0.92	0.75	0.75	0.79	0.79	0.79
ROC-AUC	0.69	0.93	0.95	0.95	0.95	0.76	0.89	0.90	0.89	0.90
PR-AUC	0.66	0.92	0.95	0.95	0.92	0.72	0.88	0.86	0.88	0.86
Log Loss	0.59	0.33	0.34	0.24	0.24	0.46	0.36	0.34	0.34	0.36
Cohen Kappa	0.31	0.75	0.79	0.84	0.84	0.53	0.53	0.59	0.59	0.59
MCC	0.42	0.75	0.80	0.85	0.84	0.60	0.60	0.63	0.63	0.63
Hamming Loss	0.34	0.13	0.11	0.08	0.08	0.23	0.23	0.20	0.20	0.20
Exec Time (s)	30.18	21.65	14.84	41.97	92.63	27.07	12.55	11.48	347.52	347.52

additional computational operations introduced by iterative learning rather than blockchain inefficiency. Storage efficiency analysis further confirms the lightweight nature of the proposed blockchain design. DDANet and RDDANet maintain highly compact chain sizes of 1197 bytes and 1199 bytes, respectively, indicating linear and controlled growth of blockchain storage without redundancy or inflation. ADDANet exhibits a higher storage footprint (6741 bytes), which is directly associated with enriched learning metadata rather than blockchain structural cost.

From a validation and scalability perspective, RDDANet achieves the lowest validation time of 0.000493 s, demonstrating the effectiveness of reinforcement learning in optimizing blockchain transaction verification. DDANet and ADDANet record 0.000853 and 0.000777 s, respectively, while still maintaining computational efficiency suitable for IoT deployment. Scalability evaluation further confirms the robustness of the blockchain layer. Under increasing IoT device loads, scalability latency remains nearly constant across all models, with RDDANet and ADDANet achieving 0.000013 s and DDANet 0.000022 s. This demonstrates that the system exhibits strong horizontal scalability with negligible marginal overhead per additional transaction, which is a critical requirement for large-scale IoT ecosystems. These results rigorously validate that the proposed blockchain-enabled framework is not only lightweight and scalable but also architecturally necessary for enforcing decentralized trust and tamper-proof data integrity. Importantly, the empirical evidence confirms that these security guarantees are achieved with negligible computational, storage, and scalability overhead, thereby justifying the inclusion of blockchain over simpler hash-chain alternatives.

3.10. Generalizability evaluation on EDGE-IIoT set and UNSW-NB15

To assess the robustness and cross-domain adaptability of the proposed models, a generalizability analysis is conducted on two benchmark datasets, namely EDGE-IIoT Set [33] and UNSW-NB15 [34]. The results are presented in Table 10, demonstrating the performance of baseline and proposed models across multiple evaluation metrics. On the EDGE-IIoT Set dataset, a significant performance improvement is observed as the models evolve from baseline to advanced architectures. The DANet achieves a relatively low accuracy of 0.66, indicating its limited capability in capturing complex feature dependencies. The DNN improves the accuracy to 0.87, highlighting the effectiveness of deep feature extraction. The proposed DDANet further enhances performance, achieving an accuracy of 0.89, which validates the benefit of integrating dense learning with dual attention mechanisms. The advanced variants, RDDANet and ADDANet, achieve the highest accuracy of 0.92, along with superior precision (0.93 and 0.92), recall (0.92), and F1-score (0.92), as shown in Table 10. These models also achieve the lowest LL (0.24) and HL (0.08), indicating reduced prediction uncertainty and

Table 11

Performance comparison of proposed DDANet, ADDANet and RDDANet models with different GAI techniques.

Metric	Generative Adversarial Network			Auto Encoder Network		
	DDANet	ADDANet	RDDANet	DDANet	ADDANet	RDDANet
Accuracy	0.84	0.88	0.90	0.82	0.85	0.87
Precision	0.83	0.87	0.89	0.81	0.84	0.86
Recall	0.82	0.86	0.88	0.80	0.83	0.85
F1-score	0.83	0.87	0.89	0.81	0.84	0.86
CK	0.72	0.77	0.79	0.70	0.74	0.76
MCC	0.71	0.76	0.78	0.69	0.73	0.75
ROC-AUC	0.88	0.89	0.90	0.86	0.88	0.89
PR-AUC	0.84	0.88	0.89	0.82	0.85	0.87
Log Loss	0.40	0.32	0.30	0.44	0.36	0.34
Hamming Loss	0.28	0.22	0.20	0.31	0.27	0.25

improved classification reliability. Furthermore, higher CK (0.84) and MCC (0.85) values demonstrate strong agreement between predicted and true labels. The ROC-AUC and PR-AUC values reaching up to 0.95 confirm excellent discriminative capability. These results collectively demonstrate that the proposed models generalize effectively on complex IoT traffic data.

On the UNSW-NB15 dataset, which introduces different traffic distributions and attack patterns, the models maintain consistent performance trends. The baseline DAN and DNN achieve an accuracy of 0.77, while DDANet improves it to 0.79. The proposed RDDANet and ADDANet achieve the highest accuracy of 0.80, as reported in Table 10. In addition, these models maintain stable precision (0.84), recall (0.80), and F1-score (0.79), indicating consistent predictive performance across datasets. The ROC-AUC values reach up to 0.90 and PR-AUC up to 0.88, demonstrating strong classification capability under domain shift conditions. Moreover, LL is reduced to 0.34 in DDANet and RDDANet, reflecting improved confidence in predictions. The improvements in CK (0.59) and MCC (0.63) further confirm reliable and stable classification performance. Although RDDANet and ADDANet incur higher execution time due to reinforcement learning and active learning mechanisms, the achieved performance gains justify the computational overhead. The results in Table 10 confirm that the proposed hybrid and learning-enhanced architectures exhibit strong generalization capability across heterogeneous datasets. The consistent improvements across all evaluation metrics indicate that the models effectively capture generalized intrusion patterns rather than overfitting to a specific dataset, making them suitable for real-world IoT intrusion detection scenarios.

3.11. Comparison of generative AI techniques for data balancing

To analyze the impact of different GAI techniques for data balancing, we compare Generative Adversarial Network (GAN) and Auto Encoder (AE) based augmentation strategies across DDANet, ADDANet, and RDDANet, as reported in Table 11. Data balancing is critical in intrusion detection because class imbalance can bias the model toward majority classes. Generative models attempt to synthesize minority class samples to improve class representation. However, the quality and distributional consistency of generated samples directly influence classification performance. Under GAN-based balancing, DDANet achieves an accuracy of 0.84, ADDANet improves to 0.88, and RDDANet reaches 0.90. Similar improvements are observed for F1-score, where RDDANet attains 0.89. Agreement metrics remain moderate, with CK values ranging from 0.72 to 0.79 and MCC values between 0.71 and 0.78. Although GAN improves minority representation, instability in adversarial training introduces synthetic noise, which slightly affects generalization. Error metrics further reflect this limitation, as LL remains relatively high (0.30 for RDDANet) and HL stays at 0.20.

In comparison, AE-based balancing yields slightly lower performance. The highest accuracy under AE is 0.87 for RDDANet, while ADDANet achieves 0.85 and DDANet records 0.82. F1-score values follow a similar pattern, reaching 0.86 for RDDANet. Agreement metrics decrease compared to GAN, with MCC values remaining below 0.75 for DDANet and below 0.76 for RDDANet. LL is still higher (0.34 when associated with RDDANet) and HL rises to 0.25. It implies that the reconstruction-based generation is unable to represent complex minority class distributions to a full extent, hence the discriminative improvement is limited.

In comparison with the already reported Real-NVP based balancing strategy, there is an apparent superiority. Real-NVP retains the probability density with invertible transformations, meaning the alignment of distribution at the real and synthetic samples is better. With this property, there is more stable and high-quality generation of samples. As the previous findings indicate, Real-NVP with ADDANet and RDDANet attains much higher accuracy (0.96), better measures of agreement (CK and MCC up to 0.95), less LL (0.12), and less HL (0.04). The flow-based architecture of Real-NVP can precisely estimate the likelihood resulting in superior modeling of minorities classes than adversarial or reconstruction-based approaches are. Thus, Real-NVP displays better stability, preservation of distribution, and improvement of classification, hence making it the best GAI method in balancing data in the proposed framework.

3.12. Comparative analysis of reinforcement learning techniques for RDDANet optimization

Table 12 of the performance comparison of various RL methods shows the effectiveness of the A2C paradigm that is used in the proposed RDDANet model. REINFORCE has the best overall performance, as it has an accuracy of 0.89, precision of 0.90, recall of 0.88, and F1-score of 0.89. It has also registered high ROC-AUC and PR-AUC of 0.91 with the lowest LL (0.29) and HL (0.13).

Table 12
Performance comparison of proposed RDDANet model with different RL techniques.

Metric	PPO	REINFORCE	DQL
Accuracy	0.82	0.89	0.88
Precision	0.83	0.90	0.89
Recall	0.75	0.88	0.87
F1-score	0.79	0.89	0.88
MCC	0.68	0.77	0.75
CK	0.66	0.76	0.74
ROC-AUC	0.86	0.91	0.90
PR-AUC	0.85	0.91	0.90
Log Loss	0.36	0.29	0.30
Hamming Loss	0.22	0.13	0.15

Table 13
Comparison of proposed ADDANet model using entropy, uncertainty, and diversity-based AL strategies.

Metric	Entropy	Uncertainty	Diversity
Accuracy	0.83	0.87	0.90
Precision	0.84	0.88	0.91
Recall	0.80	0.89	0.92
F1-score	0.82	0.88	0.91
MCC	0.68	0.76	0.81
CK	0.67	0.75	0.80
ROC-AUC	0.87	0.91	0.93
PR-AUC	0.86	0.90	0.92
Log Loss	0.35	0.29	0.26
Hamming Loss	0.21	0.15	0.11

Nonetheless, REINFORCE is a pure policy-gradient algorithm and highly sensitive to variance in gradient estimation, and is likely to exhibit unstable convergence when applied to high-dimensional intrusion detection. The DQL has competitive results with the accuracy of 0.88 and F1-score of 0.88. Its ROC-AUC and PR-AUC of 0.90, though, are with slightly higher values of its LL (0.30) and HL (0.15) than that of REINFORCE, meaning that its calibration with probability is somewhat less and that it has somewhat more near misses. DQL being a value-based approach is restricted on large or continuous action spaces and usually necessitates experience replay and target networks, making it computationally more complex.

PPO, which is also a clipped surrogate actor-to-critic strategy, performs relatively worse with the accuracy of 0.82 and F1-score of 0.79. The fact that its recall of 0.75 is less indicates that it has low sensitivity to minority attack classes. Despite the fact that PPO ensures better training stability based on constrained policy updates, the clipping mechanism can limit adequate exploration in dynamically balanced network traffic settings. This is because the proposed RDDANet uses an actor-to-critic framework which combines policy optimization (actor) and value estimation (critic), thus resulting in a decrease in gradient variance whilst sustaining consistent learning behavior. This balanced optimization accounted the high discrimination ability as indicated by higher MCC (0.77), CK (0.76) and strong ROC-AUC values. The actor-to-critic model facilitates effective reward-based adaptation and class-dependent learning and, therefore, is better adapted to detect intrusion in an imbalanced IoT setup than more basic value-based or high-variance policy-gradient models.

3.13. Comparative evaluation of active learning strategies in ADDANet

Table 13 compares the performance of the proposed ADDANet model using the entropy, Uncertainty, and diversity-based AL strategies. The findings show that the diversity-based strategy has the most successful overall performance in all the evaluation metrics. It reports an accuracy of 0.90, precision of 0.91, recall of 0.92 and F1-score of 0.91. It also shows high MCC (0.81) and CK (0.80), which proves better agreement of the classes and balanced predictive ability in an imbalanced intrusion detection situation. Entropy-based strategy, though effective in conceptual terms to measure the information content, achieves relatively low results with an accuracy of 0.83 and F1-score of 0.82. The values of its MCC (0.68) and CK (0.67) are lower, which means that its predicted and true labels are less strongly correlated. On the same note, the Uncertainty-based approach has high performance relative to Entropy with an accuracy of 0.87 and F1-score of 0.88 and ROC-AUC and PR-AUC of 0.91 and 0.90 respectively. Nonetheless, it has a slightly lower recall (0.89 vs. 0.92) and HL (0.15 vs. 0.11) performance as compared to the diversity-based strategy.

The fact that the diversity-based method has the lowest LL (0.26) and HL (0.11) also supports its superiority, which means that it has a more accurate calibration of probabilities and a small number of misclassified labels. Such enhancement can be explained by the fact that the MC-based sampling mechanism is used in the suggested framework. The Monte Carlo method approximates predictive posterior distributions by conducting a series of stochastic forward passes, and is also much more effective at representing epistemic

uncertainty. It allows the model to choose not only uncertain, but also feature space diverse samples, which reduces redundancy and increases minority attack patterns representation. The MC-based plan improves the exploration of complex IoT traffic distributions and is such that the chosen samples make the most information gain when performing incremental training. Consequently, the ADDANet model can obtain better discriminative power, generalization and robustness to class imbalance in comparison with solely Entropy or single measure Uncertainty-based AL approaches.

4. Conclusion, future work and real world application

This paper provided a thorough IoT intrusion detection model that incorporated GAI, hybrid DL, AL, RL, blockchain, statistical validation, and explainable AI. A synthetic data generation mechanism based on Real-NVP was created on the basis of GAI to solve extreme class imbalance in IoT datasets. The method enhanced the realistic representation of the minority classes and minimised bias against the majority classes by maintaining the underlying probability distribution and creating realistic samples of the minority classes, increasing stability in detection. According to the balanced dataset, DDANet was suggested as a hybrid network and incorporates a DNN and DANet. DNN identified interaction of nonlinear features globally, whereas channel and spatial attention mechanism placed focus on discriminative features, enhancing intrusion detection in high-dimensional IoT traffic. ADDANet In order to decrease the labeling cost and enhance the data efficiency, ADDANet enabled an MC-based AL strategy, which approximated the predictive uncertainty using stochastic forward pass and inferential samples, reducing redundant annotations, and preserving a high level of performance. RDDANet also incorporated an A2C reinforcement learning architecture in order to allow the adaptive optimization in dynamic IoT settings whereby the actor can learn optimal policies and the critic can stabilize the convergence by evaluating state-values. Transparency and auditability in distributed IoT systems were improved by a blockchain layer that provided secure logging and decentralized trust management and storage of detection outcomes that were tamper-resistant. The experimental findings showed apparent improvements within the proposed models. DDANet achieved 0.93 accuracy, 0.94 precision, 0.93 recall, and 0.93 F1-score, with 0.91 CK, 0.91 MCC, 0.99 ROC-AUC, 0.98 PR-AUC, 0.21 Log Loss, 0.07 Hamming Loss, and 21.96 s execution time. ADDANet improved performance to 0.96 accuracy, 0.96 precision, 0.96 recall, and 0.96 F1-score, with 0.94 CK, 0.94 MCC, 1.00 ROC-AUC, 0.99 PR-AUC, 0.13 Log Loss, 0.04 Hamming Loss, and 132.69 s execution time. RDDANet maintained 0.96 accuracy, 0.96 precision, 0.96 recall, and 0.96 F1-score, with enhanced 0.95 CK and 0.95 MCC, 0.99 ROC-AUC, 0.99 PR-AUC, the lowest Log Loss of 0.12, 0.04 Hamming Loss, and 52.36 s execution time. Robustness and generalization were validated using 10-FCV, reducing overfitting and ensuring stable performance across folds. Confidence interval analysis confirmed reliability, and paired *t*-test results verified statistical significance. Finally, IG-based explainable AI provided feature attribution scores, enhancing interpretability and trust. The integration of GAI, hybrid attention modeling, active and reinforcement learning, blockchain security, and statistical validation established an adaptive, secure, and high-performing IoT intrusion detection framework.

Future work will focus on extending the proposed framework toward broader scalability, adaptability, and real-world deployment. Advanced RL strategies and multi-agent optimization mechanisms can further enhance adaptability in highly dynamic and large-scale IoT environments. The blockchain layer can also be extended by incorporating automated response mechanisms for real-time mitigation of detected intrusions. Furthermore, the proposed framework will be validated on additional publicly available IoT intrusion detection datasets to assess its generalization capability, robustness across diverse traffic patterns, and effectiveness against varying attack distributions.

The proposed framework can be applied in a wide range of real-world IoT environments where security, adaptability, and trust are essential. In smart healthcare, it can protect connected medical devices and patient-monitoring systems from web-based and network intrusion attacks; in smart transportation and intelligent traffic systems, it can detect malicious traffic patterns that may disrupt connected vehicles or roadside infrastructure; in smart grids and industrial control systems, it can support early intrusion detection while ensuring secure and tamper-resistant logging of security events through blockchain integration. The framework is also well suited for large-scale edge-cloud IoT deployments, where evolving attack behavior, data imbalance, and the need for explainable decisions require adaptive learning, efficient sample selection, and transparent security monitoring. These application domains align with the paper's focus on dynamic IoT environments, secure distributed systems, and high-performance intrusion detection with explainability and decentralized trust.

CRedit authorship contribution statement

Zahoor Ali Khan: Writing – review & editing, Formal analysis, Data curation, Conceptualization; **Ifra Shaheen:** Writing – original draft, Visualization, Validation, Methodology; **Nadeem Javaid:** Writing – review & editing, Supervision, Investigation, Conceptualization; **Imran Ahmed:** Visualization, Formal analysis, Data curation, Conceptualization; **Muhammad Shafiq:** Resources, Investigation, Funding acquisition, Formal analysis; **Jin-Ghoo Choi:** Writing – review & editing, Visualization, Formal analysis, Data curation.

Data availability

We have shared the link of dataset in the manuscript

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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